

Computer Networks

Medium Access Sublayer

Here we want to know how to handle broadcast networks. As compared to point to point networks, a major issue is handling arbitration when there is competition for the network.

This is the bottom sublayer of the Data Link Layer.

The Channel Allocation Problem

How to allocate a single channel among multiple users.

Multiple Access Protocols

How to handle contention for the use of a channel.

Bridges

Ways of connecting networks together.

High Speed LANs

Directions in high speed networks.

CHANNEL ALLOCATION PROBLEM:

STATIC CHANNEL ALLOCATION IN LANs AND MANS

The traditional (phone company) way of allocating a single channel is Frequency Division Multiplexing. (See Figure) FDM works fine for limited and fixed number of users.

Inefficient to divide into fixed number of chunks. May not all be used, or may need more. Doesn't handle burstiness.

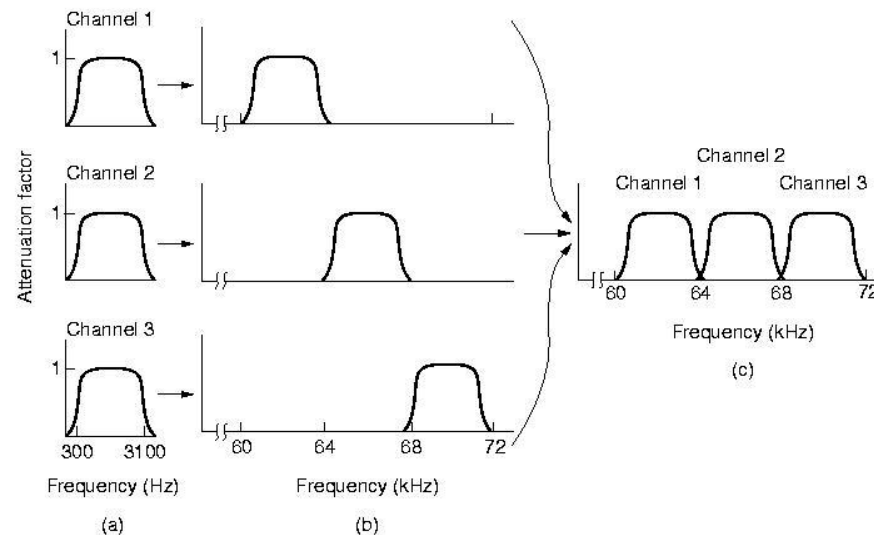
T = mean time delay

C = capacity (bps)

λ = arrival rate

$1/\mu$ = mean length

$$T = \frac{1}{\mu C - \lambda}$$



CHANNEL ALLOCATION PROBLEM:

STATIC CHANNEL ALLOCATION IN LANs AND MANS

Now divide this channel into N subchannels, each with capacity C/N . Input rate on each of the N channels is λ / N . Now

$$T(\text{fdm}) = \frac{1}{\mu(C/N) - \lambda / N} = \frac{N}{\mu C - \lambda} = NT$$

The mean time delay using FDM is N times worse if all the frames were somehow magically arranged orderly in a big central queue.

CHANNEL ALLOCATION PROBLEM:

DYNAMIC CHANNEL ALLOCATION

Possible underlying assumptions include:

Station Model -

Assumes that each of N "stations" (packet generators) independently produce frames. The probability of producing a packet in the interval dt is λdt where λ is the constant arrival rate. That station generates no new frame until that previous one is transmitted.

Single Channel Assumption -

There's only one channel; all stations are equivalent and can send and receive on that channel.

Collision Assumption -

If two frames overlap in any way time-wise, then that's a collision. Any collision is an error, and both frames must be retransmitted. Collisions are the only possible error.

CHANNEL ALLOCATION PROBLEM:

DYNAMIC CHANNEL ALLOCATION

Continuous Time -

There's no "big clock in the sky" governing transmission. Time is not in discrete chunks.

Slotted Time -

Alternatively, frame transmissions always begin at the start of a time slot. Any station can transmit in any slot (with a possible collision.)

Carrier Sense -

Stations can tell a channel is busy before they try it. NOTE - this doesn't stop collisions. LANs have this, satellite networks don't.

Multiple Access Protocols

Many algorithms for allocating a multiple access channel are known.

ALOHA:

Developed in Hawaii in the 1970s for packet radio network. It is applicable to any shared medium.

PURE ALOHA:

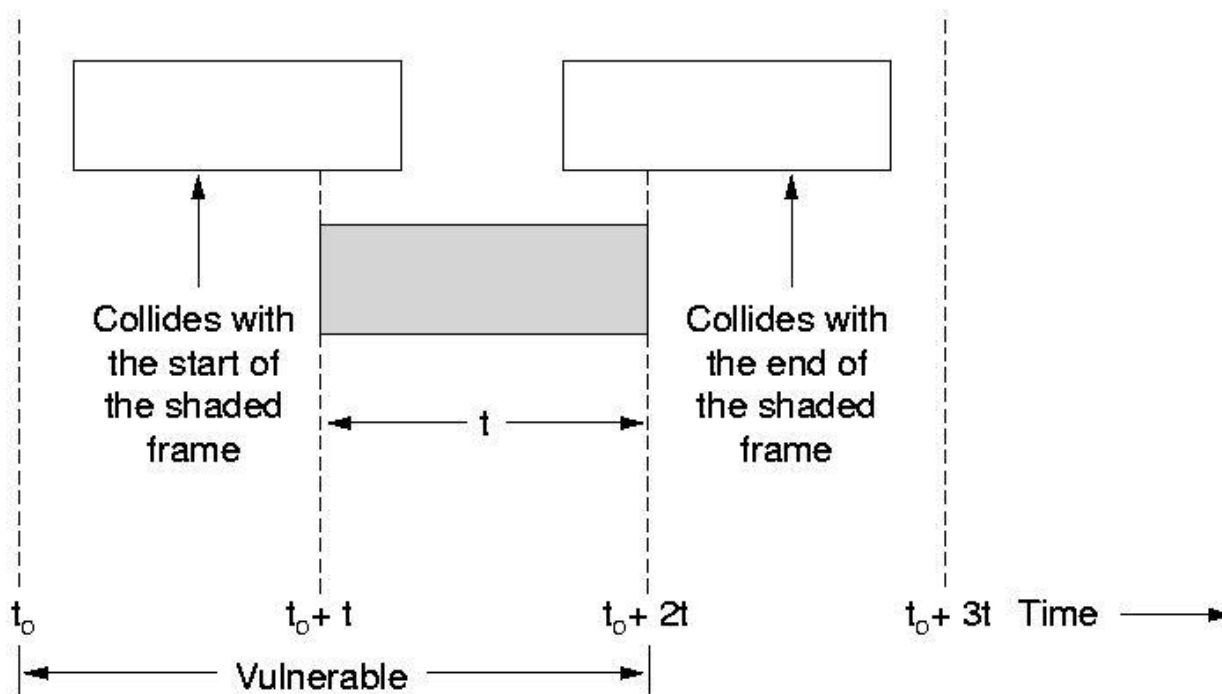
Every station transmits whenever it wants to. Colliding frames are destroyed. The sender knows if its frame got destroyed, and if so waits a random time and then retransmits.

- When station has frame, it sends
- Station listens (for max round trip time) plus small increment
- If ACK, fine. If not, retransmit
- If no ACK after repeated transmissions, give up
- If frame OK and address matches receiver, send ACK
- Frame may be damaged by noise or by another station transmitting at the same time (collision)
- Any overlap of frames causes collision
- Max utilization 18%
- Best efficiency if frames are same size.

Efficiency of Pure Aloha System

- S = frames to be transmitted. In units of frames per frame time. (What is the meaning of frame time as used here??)
- Mean $G = S$ + frames retransmitted due to previous collisions.
- P_0 = probability that a frame does NOT suffer collision.
- Throughput S is just the offered load, G times the probability, P_0 , of a transmission succeeding that is $S = P_0 \times G$

Use the Figure to determine collision vulnerability.



Probability that k frames are generated during a given frame time (Poisson distribution):

$$\Pr[k] = \frac{G^k e^{-G}}{k!}$$

So the probability of zero frames generated during frame time = e^{-G} .

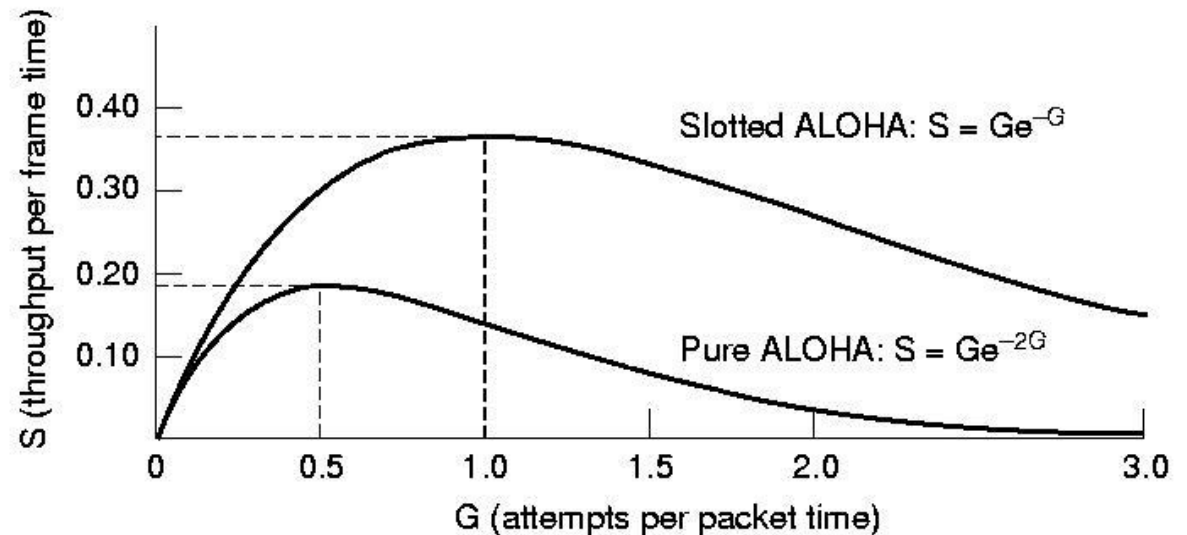
In the interval two frame time long, the mean number of frame generated is $2G$.

The Probability of no other traffic being initiated during the vulnerable period:

$$P_0 = e^{-2G} \quad \text{so}$$

Throughput per frame time is:

$$S = G e^{-2G}$$



Slotted Aloha System

- Time in uniform slots equal to frame transmission time
- Need central clock (or other sync mechanism)
- Transmission begins at slot boundary
- Frames either miss or overlap totally
- Max utilization 37%

Doubles efficiency by dividing time into "ticks". Sends occur only at the start of a tick. Vulnerable period is 1/2 of pure Aloha case, so

$$S = G e^{-G}$$

Slotted aloha peaks at $G = 1$, with a throughput of $S = 1/e$ or about .368 (.37), twice of pure Aloha.

If the system is operating at $G=1$, the probability of empty slot is 37.

Best throughput is at $G = 1$ when

$$\text{success } S = 0.37; \text{ empty slots} = \Pr[0] = 0.37; \text{ collisions} = 0.26$$

CARRIER SENSE MULTIPLE ACCESS PROTOCOLS

The protocols in which stations listen for a carrier (a transmission) and act accordingly, are called carrier sense protocol.

CSMA:

- All stations know that a transmission has started almost immediately
- First listen for clear medium (carrier sense)
- If medium idle, transmit
- If two stations start at the same instant, collision
- Wait reasonable time (round trip plus ACK contention)
- No ACK then retransmit
- Max utilization depends on propagation time (medium length) and frame length
- Longer frame and shorter propagation gives better utilization

There are four type of protocols based on CSMA.

- 1-PERSISTENT CSMA
- 0-PERSISTENT CSMA
- p-PERSISTENT CSMA
- CSMA with CD

1-PERSISTENT CSMA Protocol:

To avoid idle channel time, 1-persistent protocol used.

1. **Station listens. If channel idle, it transmits; otherwise go to step 2.**
 2. **If channel busy, continue to listen until the channel is sensed idle. Then transmit immediately.**
- if two or more stations are waiting to transmit, a collision is guaranteed. If collision, wait a random time and try again
 - 1-persistent stations selfish

Nonpersistent CSMA Protocol (0-persistent CSMA)

1. If medium is idle, transmit; otherwise, go to 2
 2. If medium is busy, wait amount of time drawn from probability distribution (retransmission delay) and repeat 1
- if two or more stations are waiting to transmit at the same time, a collision may occur. If collision, wait a random time and try again
 - Random delays reduces probability of collisions
 - Consider two stations become ready to transmit at same time
 - While another transmission is in progress
 - If both stations delay same time before retrying, both will attempt to transmit at same time
 - Capacity is wasted because medium will remain idle following end of transmission
 - Even if one or more stations waiting
 - Nonpersistent stations deferential

Leads to

- 1) better utilization and
- 2) longer delays than 1 - persistent. (why?)

p-Persistent CSMA Protocol

Compromise that attempts to reduce collisions

- Like nonpersistent
- And reduce idle time
 - Like 1-persistent
- **Rules:**
 1. **If medium idle, transmit with probability p , and delay one time unit with probability $(1 - p)$**
 - Time unit typically maximum propagation delay
 2. **If medium busy, listen until idle and repeat step 1**
 3. **If transmission is delayed one time unit, repeat step 1**
- if two or more stations are waiting to transmit at the same time, a collision may occur. If collision, wait a random time and try again.
- **What is the effective value of p ?**
- If heavy load expected, p must be small. However, as p made smaller, stations wait longer but reduces the collisions.
- At low loads, this gives very long delays.
- Eg. At low load, if $p = 0.1$, a station will wait an average of 9 time units before transmitting on an idle line.

CSMA WITH COLLISIONS DETECTION (CSMA/CD)

When two frame collide, the medium remains unusable for the duration of transmission of both frames.

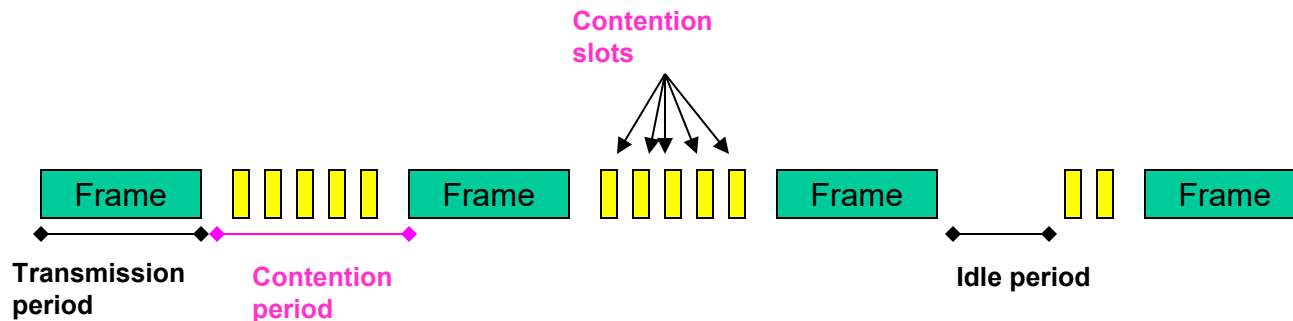
For long frames, compared to propagation time, the amount of wasted capacity can be considerable.

The waste can be reduced if a station continues to listen to the medium while transmitting.

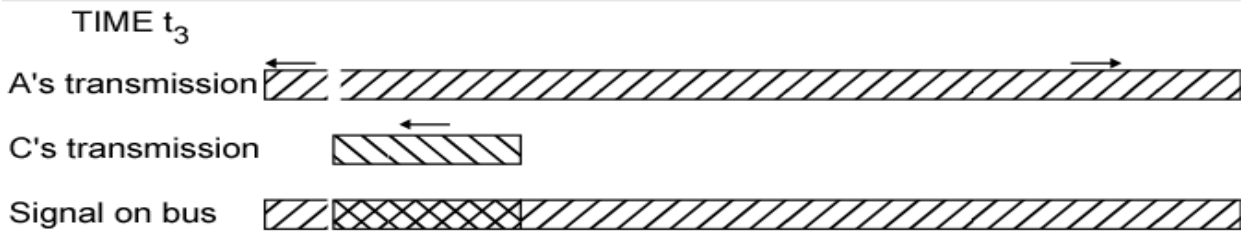
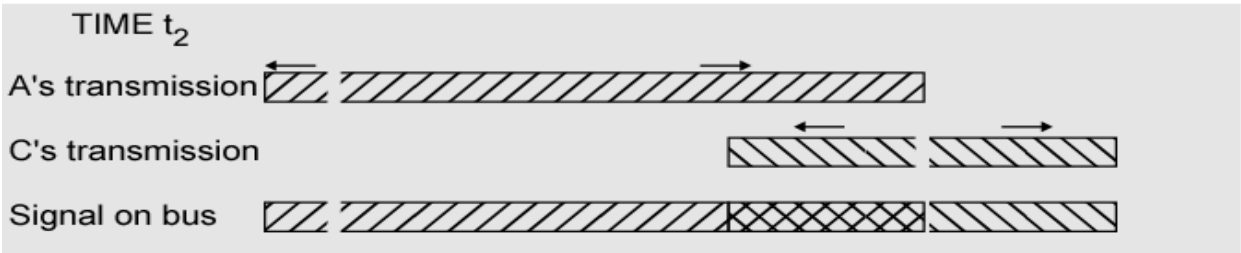
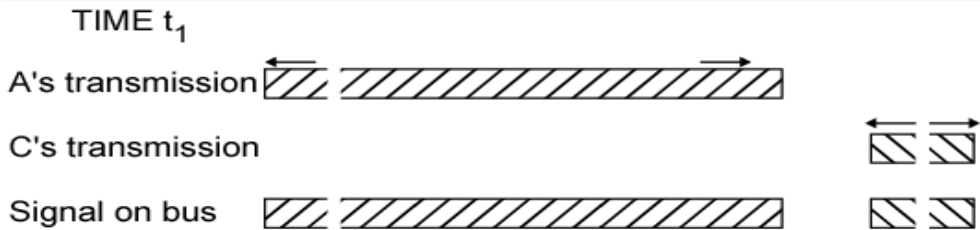
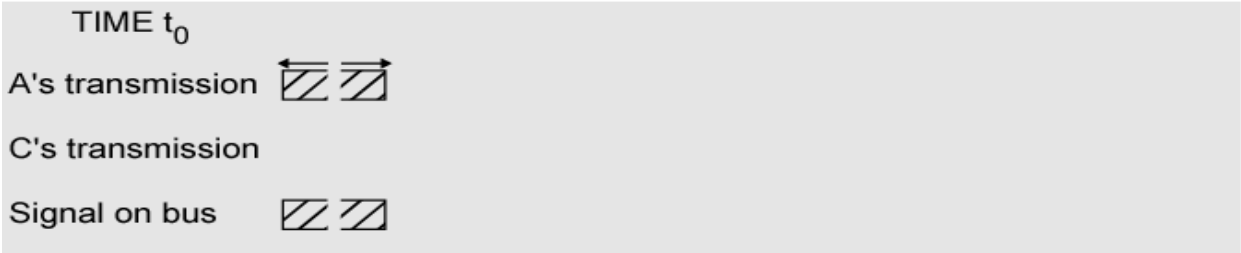
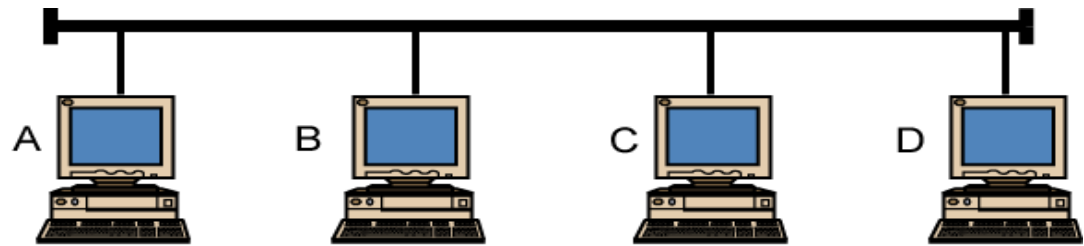
This leads to the following rules:

1. **If medium idle, transmit, otherwise, step 2**
2. **If the medium is busy, continue to listen until the channel is sensed idle, then transmit**
3. **If collision detected during transmission, it stops sending, even if in mid-frame.**
4. **Wait a random amount of time then start from step 1**

What is contention interval -- how long must station wait after it sends until it knows it got control of the channel? It's twice the time to travel to the furthest station.



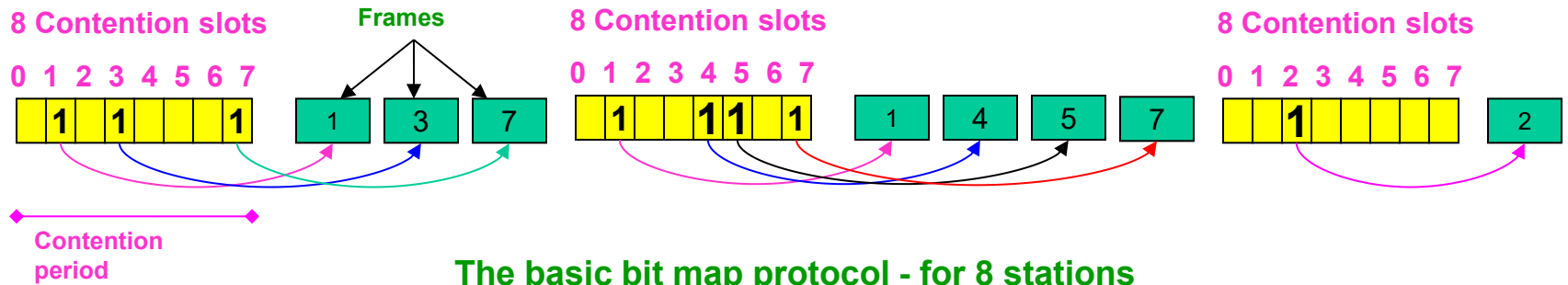
CSMA/CD Operation



COLLISION-FREE PROTOCOLS

Bit Map Protocol

- There are N stations, each with a unique address from 0 to N-1.
- Contention period is divided into exactly N slots.
- If station 0 has a frame to send, it transmits a 1 bit during the zeroth slot. No other station is allowed to transmit during this slot.
- In general, station j may announce that it has a frame to send by inserting a 1 bit into slot j.
- After the announcement, they begin transmitting in numerical order,
- There will be never collisions.



What are pros and cons of this method? Analyze at low and high loads.

Performance Analysis of Bit Map Protocol

Let size of data frame is d bits and each contention slot is 1 bit long.

At low load:

Bit map will be repeated over and over. Consider the situation for low-numbered (0,1) and high numbered stations.

Low numbered:

Let 0 or 1 station is ready when current scan is some where in the middle of the bit-map.
Waiting time to start transmission = $N/2 + N = 1.5 N$

For high numbered: ready when current scan is some where in the middle of the bit-map.
waiting time = $N/2$
Mean waiting time = $(1.5N + .5N)/2 = N$.

So there is N bit overhead per frame. Efficiency: $d / (N + d)$

At high load:

all stations have data to send all the time. In N bit contentions, send N frames. So there is 1 bit overhead per frame.

Efficiency: $d / (d + 1)$

Question: How long does a station wait s , have to wait in the worst case before it can start transmitting its frame over a LAN that uses the basic bitmap protocol?

The worst case is: all stations want to send and s is the lowest numbered station.
Wait time N bit contention period + $(N - 1) \times d$ bit for transmission of frames.
The total is $N + (N - 1)d$ bit times.

Binary Countdown Protocol

- Each station has equal length binary address.
- Stations who willing to use the channel, broadcast their addresses from high order bit.
- Bits in each address position from different stations are OR together.
- As soon as a station sees that a high-order bit position that is 0 in it address has been overwritten with a 1, it gives up.

Example:

- If stations 0010, 0100, 1001, and 1010 all trying to get the channel.
- In first bit time, the stations transmit 0, 0, 1, and 1 respectively. Then ORed together to form a 1.
- Stations 0010, 0100 see this 1 and give up, because higher number station are competing.
- Stations 1001 and 1010 continue.
- Next bit 0, so both continue
- Next bit is 1, so station 1001 gives up.
- Station 1010 get the channel.

Channel efficiency is : $d / (d + \log_2 N)$

N- number of the maximum stations

	Bit time			
	0	1	2	3
0 0 1 0	0	-	-	-
0 1 0 0	0	-	-	-
1 0 0 1	1	0	0	-
1 0 1 0	1	0	1	0

result **1 0 1 0**

Stations 0010 and
0100 see this 1 and
give up

Stations 1001 sees
this 1 and give up

Adaptive Tree Walk Protocol

- Stations are at leaves of binary tree.
- All stations are trying to acquire the channel during slot 1. if one of them does so fine.
- If there is a collision, then during slot 1 only those stations falling under node 2 in the tree may complete.
- During slot 2, if two or more stations (under node 2) are trying to get channel, there will be collision during slot 2. then see node 4 else turn to node 3.
- If collision during slot 3, repeat it in similar way.
- Something like pre-order traversal.

Example :

Sixteen stations, numbered 1 through 16, are contending for the use of a shared channel by using the adaptive tree walk protocol. If all the station whose address are prime numbers suddenly become ready at once, how many bit slots are needed to resolve the contention?

Stations 2, 3, 5, 7, 11, and 13 want to send. Eleven slots are needed, with the contents of each slot being as follows:

slot 1: 2, 3, 5, 7, 11, 13

slot 2: 2, 3, 5, 7

slot 3: 2, 3

slot 4: 2

slot 5: 3

slot 6: 5, 7

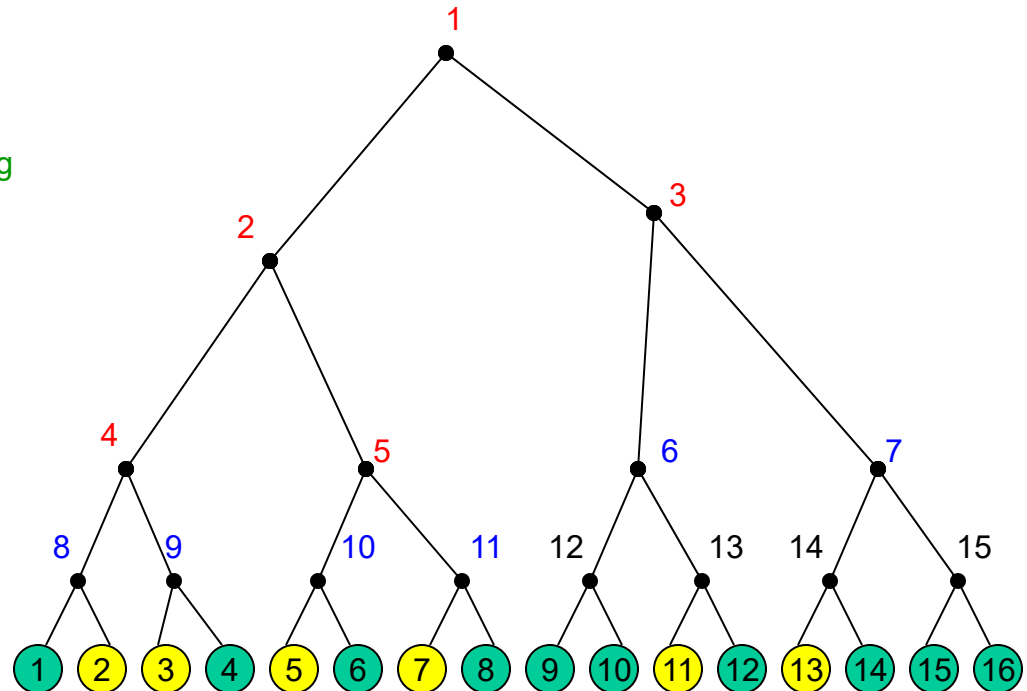
slot 7: 5

slot 8: 7

slot 9: 11, 13

slot 10: 11

slot 11: 13



Wireless LAN Protocols IEEE 802.11

Hidden Station Problem

The problem of a station not being able to detect a potential competitor for the medium because the competitor is too far away.

Example of the problem:

- A and C send a signal to B.
- But A and C aren't aware of each other's signals.
- Signals collide at B.
- But A and C don't know they collided so don't go into collision avoidance.

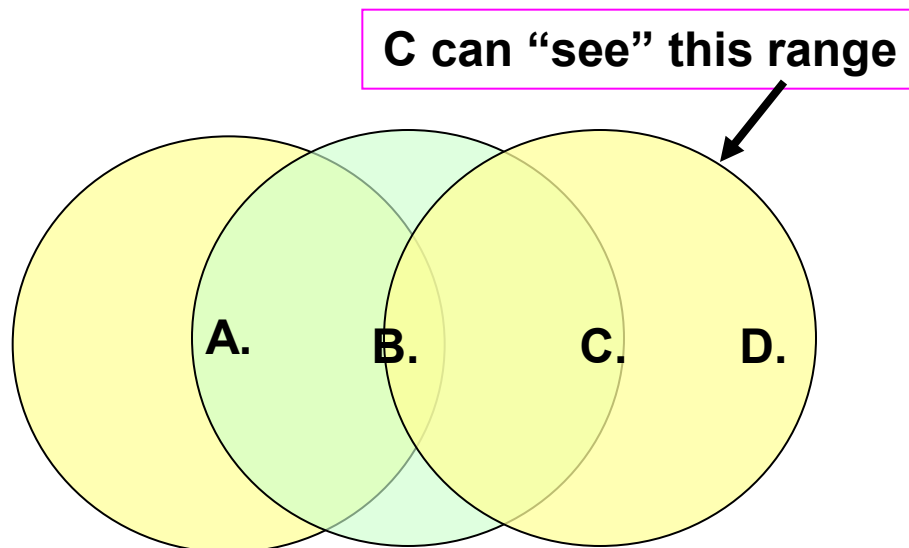
A and C are "hidden nodes"

Exposed Station Problem

Example of another problem:

- B is sending to A.
- C can hear this signal from B
- C assumes it can NOT transmit
- But C could in fact transmit to D

This is called the "exposed node problem"



Multiple Access with Collision Avoidance (MACA)

Collision Avoidance:

Problem is solved by using a protocol **Multiple Access with Collision Avoidance (MACA)**.

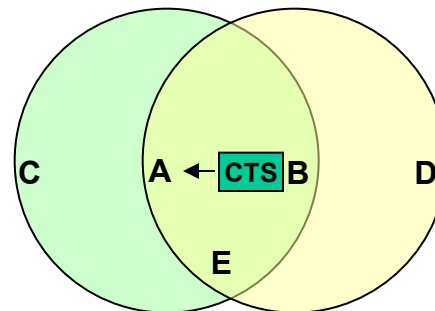
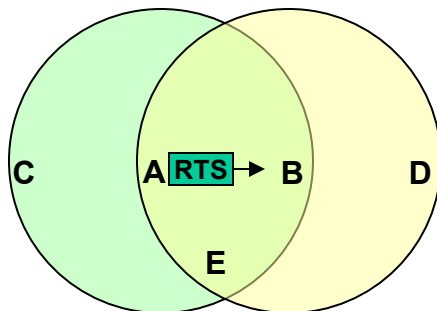
Sender and receiver exchange control frames so the transaction goes like this:

- Sender (A) does Request to Send (RTS) to receiver(B)
- B sends back Clear to Send (CTS)
- A sends packet.
- B sends an ACK after receiving the frame.

Logic:

- If a node hears the CTS, it knows it is near the receiver - so don't transmit up to the upcoming data.
- If a node hears the RTS is near the sender, must remain silent long enough for the CTS to be transmitted back to sender.
- If a node hears the RTS but not the CTS, it's not near the receiver so it can start its own transaction.

RTS and CTS contain length of packet to be sent so others know how long to wait.



BRIDGES

Bridges operate in the data link layer, and used to connect different LANs.

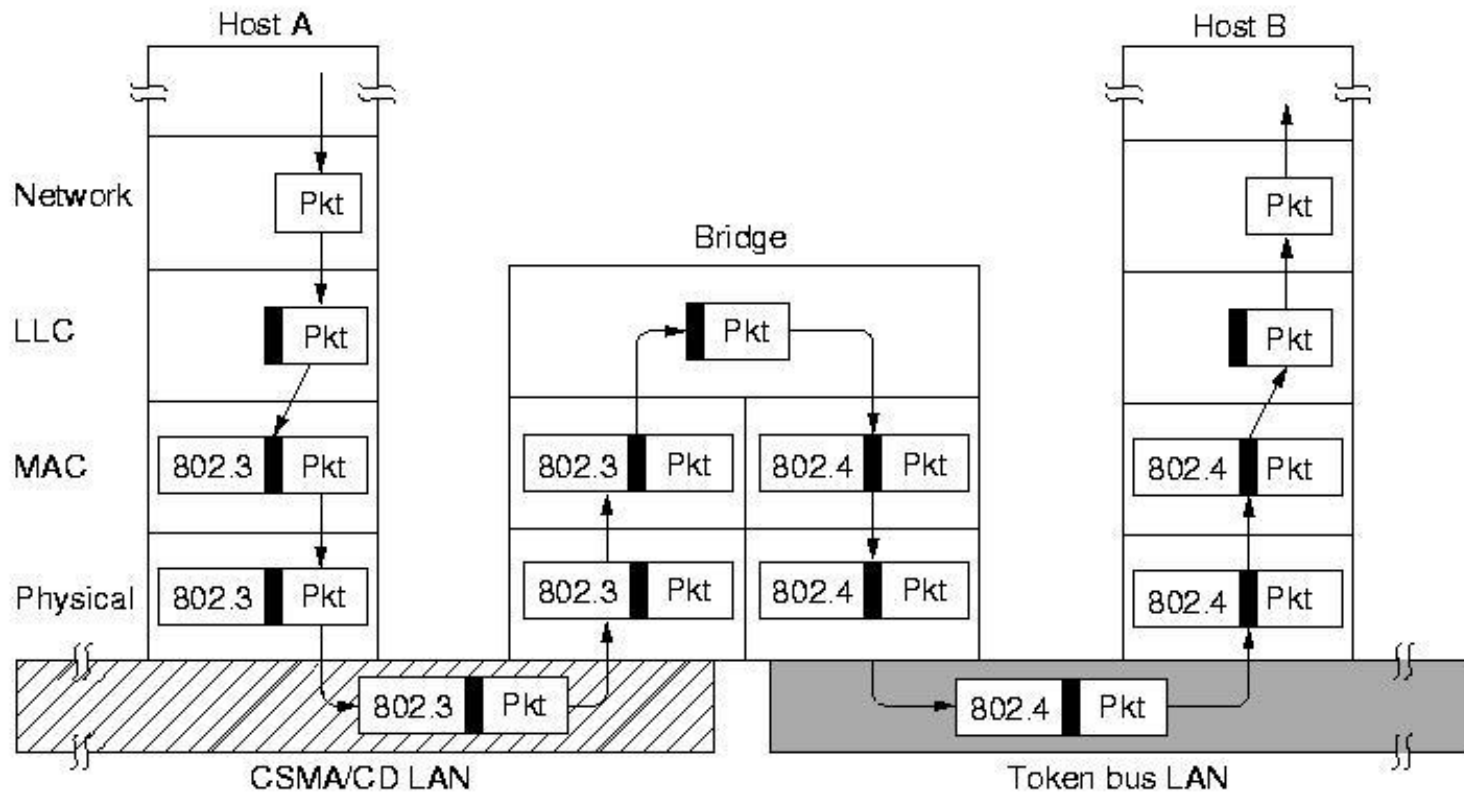
What we will talk about here -

Translation from one LAN type to another. Given a MAC address, how does a packet forwarded to its destination.

- Issues -
- Each LAN type has its own frame format. Bridges take off one type and put on another.
- LANs don't necessarily run at the same speed, so must reject or buffer the data
- Two input LANs feeding one output LAN.
- Each LAN type has it's own maximum data length. So bridges must do framing in order to translate.
- Network layers may time out because they expect the destination to ACK within a given time; all this translating slows down the transmission.
- All LAN types don't carry the same information:
 - o Priority bits.
 - o Acknowledgment bits.
- In essence, the LAN standards are incompatible.
- Total distance more than 2.5 Km. There is a problem of round-trip delay here.
- Bridges can act as firewalls, to partition against errant hardware.
- LANs broadcast everything on the LAN to all stations. May want to prevent this from happening for some data. A bridge partitions off these messages.

BRIDGES

How they work -



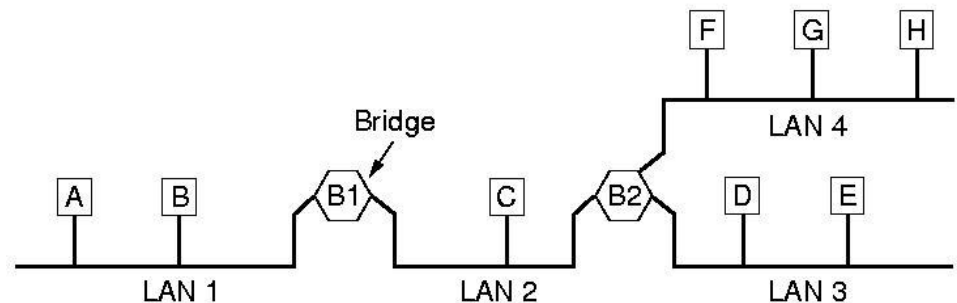
TRANSPARENT BRIDGES

A transparent bridge operate in promiscuous mode.

Promiscuous mode:

- Accepts all packets from all LANs attached to the bridge.
 - If destination is on incoming LAN, discard the packet.
 - Otherwise, forward the packet.
 - Use table (hashed) in bridge to determine choice of the LAN for forwarding.
 - Each of the hash table have : destination address and output LAN.
-
- Initially all the hash tables are empty. None of the bridges know where the destinations.
 - They uses the **flooding algorithm**: every incoming frame for an unknown destination is forward on all the LANs to which bridge is connected except the one it arrived on.
 - Transparent bridges uses **backward learning**. By looking at the source address, they learn which machine is belong to which LAN.

Example: If bridge B1 sees a frame on LAN2 coming from C. so it makes an entry in its hash table noting that frames going to C should uses LAN2.



TRANSPARENT BRIDGES

What happens if a machine is down or moved from one LAN to another or topology changed?

- Whenever a hash table entry is made, the arrival time of frame is noted in to entry or updated if entry is already there.
- The bridge periodically scan the hash table and purges all entries more than a few minutes old.

Routing Procedure:

- If destination and source LANs are the same, discard the frame.
- If destination and source LANs are different, forward the frame.
- If the destination LAN is unknown, use flooding.

Spanning Tree Bridges:

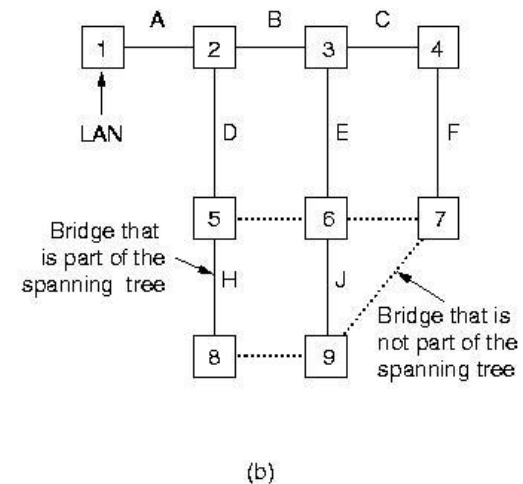
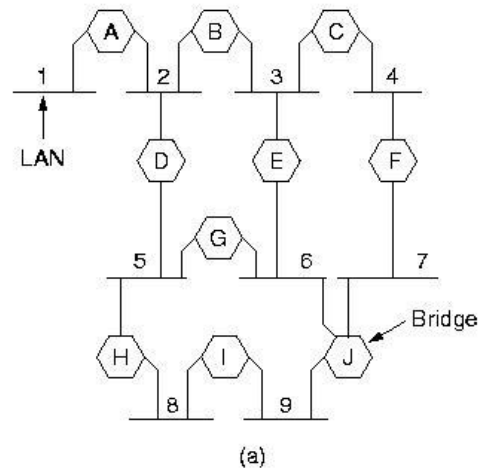
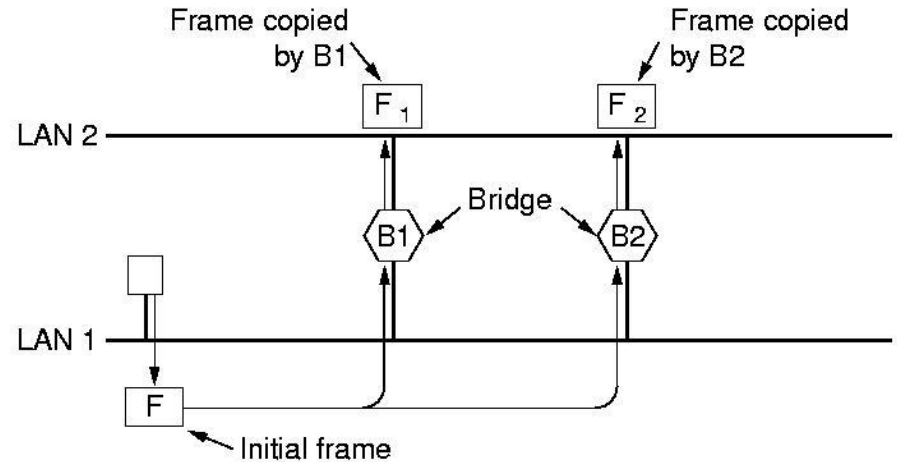
Here two or more bridges are used for reliability.

Problems with infinite flooding.

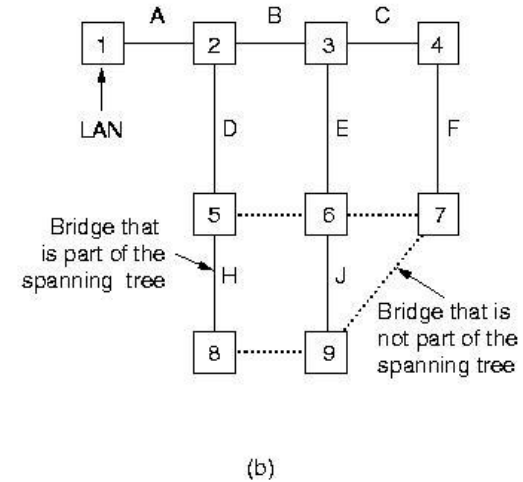
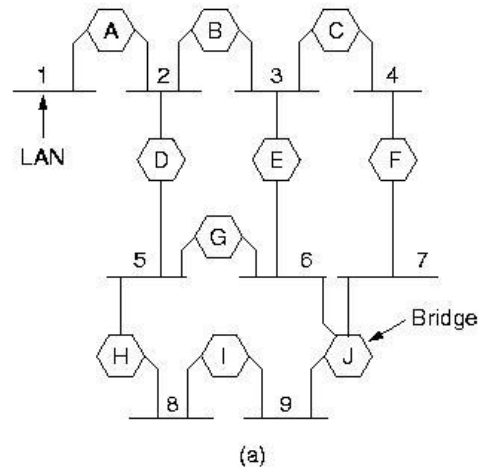
Solution is to overlay the Backward Learning policy with a virtual loop-free topology.

A Spanning Tree Bridge does this.

Note how paths are reduced.



Spanning Tree Bridges:

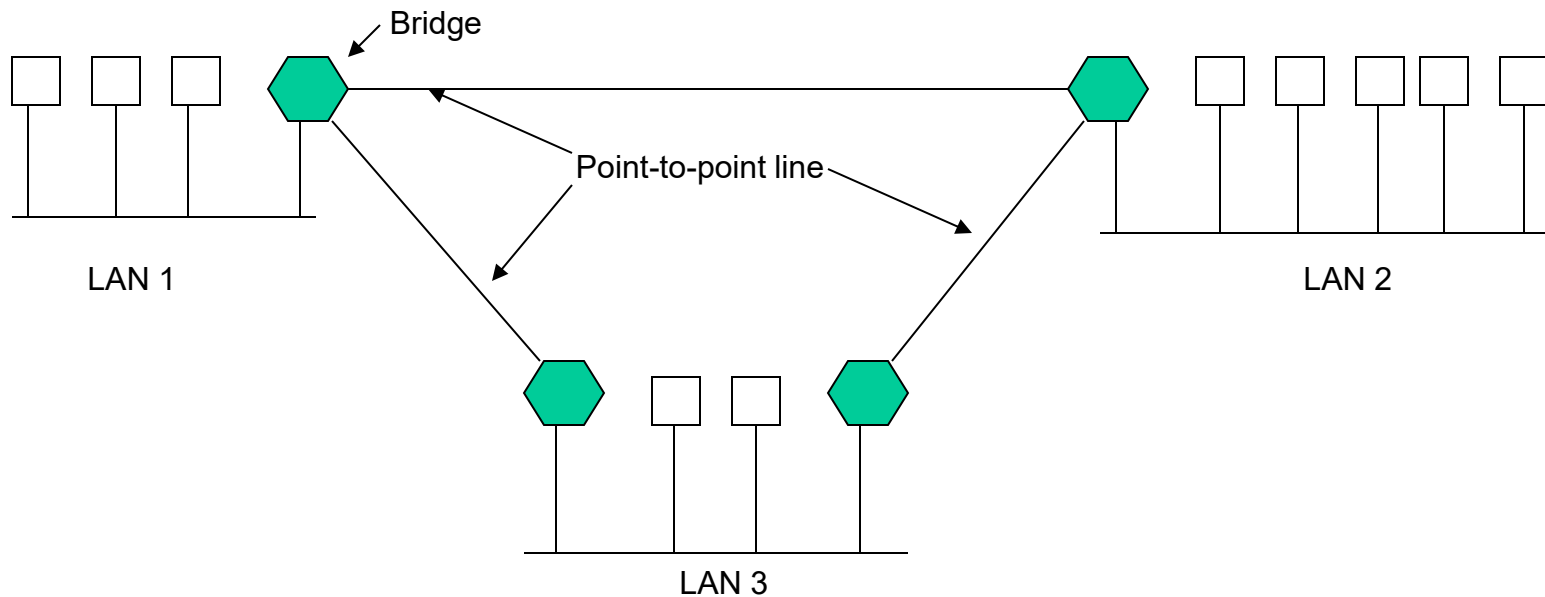


Algorithm is as follows:

- Choose one bridge to be root of tree (lowest unique serial number wins.)
- A tree for shortest paths from the root to every bridge and LAN is constructed.
- Tree includes every LAN but not necessarily every bridge.
- Continues to check for topology changes.

Remote Bridge

- The remote bridge is use to connect two or more distant LANs.
- Bridges are connected pair wise with point-to-point line.



Repeaters, Hub, Bridge, Switch, Router and Gateways

Application Gateways Application. Layer
Transport gateways Transport. Layer
Routers Network Layer
Bridge, Switches Data Link Layer
Repeaters, Hub Physical Layer

- Repeaters are analog devices that are connected to two cable segment and amplified the incoming signal.
- A hub has a number of input lines that it joins electrically. Frame are arriving on any lines are sent put on all the others. It has a single collision domain.
- Difference between bridge and switch is that when source and destination LANs are same, the frame is not discarded in switch. They have multiple collision domain (one for each line).
- Routers are used to find the best route for forwarding.
- Transport gateways use to connect two computers that uses different connection-oriented protocols. (TCP/IP and ATM Transport protocol)
- Application gateways understand the format and contents of data and translate message from one format to another.

IEEE Standards for LANs

IEEE Standards for LANs

How do the protocols of the last sections apply to real systems. Here we talk about the actual standards in use.

802.3 - Ethernet

802.4 -Token Bus

802.5 -Token Ring